ALGEBRA

Quadratic Equation

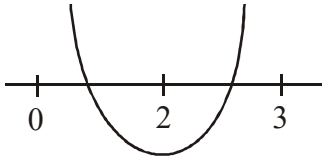
1. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0, 2) and its other root lies in the interval (2, 3). Then the number of elements in S is :

द्विघातीय समीकरण $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$ पर विचार कीजिए। माना S, c के उन सभी पूर्णांकीय मानों, जिनके लिए समीकरण का एक मूल अंतराल (0, 2) में है तथा इसका दूसरा मूल अंतराल (2, 3) में है, का समुच्चय है, तो S के अवयवों की संख्या है :

(A) 11 (B) 10 (C) 12 (D) 18

Ans. A

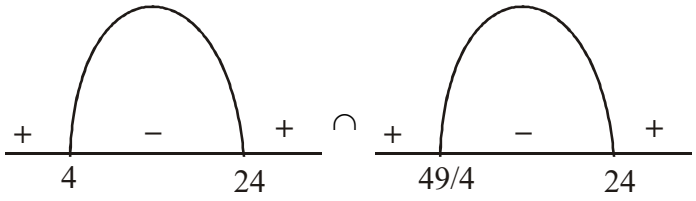
Sol.



$f(x) = (c - 5)x^2 - 2cx + c - 4$

$f(0) \cdot f(2) < 0$ ..... (1) & $f(2) \cdot f(3) < 0$

$(c - 4)(c - 24) < 0$ & $(c - 24)(4c - 49) < 0$



$c \in \left(\frac{49}{4}, 24\right) = 13, 14, \ldots\ldots 23$

No of I = 11

2. If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$ is :

(A) $2 - \sqrt{3}$ (B) $-2 + \sqrt{2}$ (C) $4 - 3\sqrt{2}$ (D) $4 - 2\sqrt{3}$

यदि x में द्विघात समीकरण $3m^2x^2 + m(m - 4)x + 2 = 0$ के मूलों का अनुपात λ है, तो m का वह न्यूनतम मान जिसके लिए $\lambda + \frac{1}{\lambda} = 1$ है, है –

(A) $2 - \sqrt{3}$ (B) $-2 + \sqrt{2}$ (C) $4 - 3\sqrt{2}$ (D) $4 - 2\sqrt{3}$

Ans. C

Sol. $3m^2x^2 + m(m - 4)x + 2 = 0$

Ratio of roots $\Rightarrow \frac{\alpha}{\beta} = \lambda$

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=1$$

$$=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=1$$

$$=\left(\frac{m(m-4)}{3m^2}\right)^2-\frac{4}{3m^2}=\frac{2}{3m^2}$$

$= m^2(m-4)^2 - 6.3\ m^2 = 0$

$= (m-4)^2 - 18 = 0$

$m = 4 - 3\sqrt{2}$

3. If m is chosen in the quadratic equation $(m^2+1)x^2 - 3x + (m^2+1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is:

(A) $8\sqrt{3}$ (B) $10\sqrt{5}$ (C) $4\sqrt{3}$ (D) $8\sqrt{5}$

यदि द्विघातीय समीकरण $(m^2+1)x^2 - 3x + (m^2+1)^2 = 0$ में m इस प्रकार लिया गया है, कि इसके मूलों का योगफल अधिकतम है, तो इसके मूलों के घन का निरपेक्ष अन्तर है :

(A) $8\sqrt{3}$ (B) $10\sqrt{5}$ (C) $4\sqrt{3}$ (D) $8\sqrt{5}$

Ans. D

Sol. $\text{Sum of roots} = \frac{3}{m^2+1}$

For maximum $m = 0$

Hence equation becomes $x^2 - 3x + 1 = 0$

$\alpha+\beta = 3,\ \alpha\beta = 1,\ |\alpha-\beta| = \sqrt{5}$

$|\alpha^3-\beta^3| = (\alpha-\beta)(\alpha^2+\beta^2+\alpha\beta)| = \sqrt{5}(9-1) = 8\sqrt{5}$

4. If α and β are the roots of the quadratic equation, $x^2 + x\sin\theta - 2\sin\theta = 0,\ \theta \in \left(0, \frac{\pi}{2}\right)$, then

$\frac{\alpha^{12}+\beta^{12}}{(\alpha^{-12}+\beta^{-12})\cdot(\alpha-\beta)^{24}}$ is equal to :

(A) $\frac{2^{12}}{(\sin\theta-8)^6}$ (B) $\frac{2^{12}}{(\sin\theta-4)^{12}}$ (C) $\frac{2^6}{(\sin\theta+8)^{12}}$ (D) $\frac{2^{12}}{(\sin\theta+8)^{12}}$

यदि द्विघात समीकरण, $x^2 + x\sin\theta - 2\sin\theta = 0,\ \theta \in \left(0, \frac{\pi}{2}\right)$ के मूल α तथा β हैं, तो

$\frac{\alpha^{12}+\beta^{12}}{(\alpha^{-12}+\beta^{-12})\cdot(\alpha-\beta)^{24}}$ बराबर है :

(A) $\frac{2^{12}}{(\sin\theta-8)^6}$ (B) $\frac{2^{12}}{(\sin\theta-4)^{12}}$ (C) $\frac{2^6}{(\sin\theta+8)^{12}}$ (D) $\frac{2^{12}}{(\sin\theta+8)^{12}}$

Ans. D

Sol. Given $\alpha+\beta=-\sin\theta$ and $\alpha\beta=-2\sin\theta$

$$\frac{\left(\alpha^{12}+\beta^{12}\right)\alpha^{12}\beta^{12}}{\left(\alpha^{12}+\beta^{12}\right)(\alpha-\beta)^{24}}=\frac{(\alpha\beta)^{12}}{(\alpha-\beta)^{24}}$$

$$|\alpha-\beta|=\sqrt{(\alpha+\beta)^2-4\alpha\beta}=\sqrt{\sin^2\theta+8\sin\theta}$$

Hence required quantity

$$\frac{(\alpha\beta)^{12}}{(\alpha-\beta)^{24}}=\frac{(2\sin\theta)^{12}}{\sin^{12}\theta(\sin\theta+8)^{12}}=\frac{2^{12}}{(\sin\theta+8)^{12}}$$

5. If α, β and γ are three consecutive terms of a non constant G.P. such that the equations $\alpha x^2+2\beta x+\gamma=0$ and $x^2+x-1=0$ have a common root then $\alpha(\beta+\gamma)$ is equal to

(A) 0 (B) $\alpha\gamma$ (C) $\beta\gamma$ (D) $\alpha\beta$

यदि एक भिन्न पदों वाली गुणोत्तर श्रेढ़ी के तीन क्रमागत पद α, β तथा γ इस प्रकार हैं कि समीकरणों $\alpha x^2+2\beta x+\gamma=0$ तथा $x^2+x-1=0$ का एक मूल समान है, तो $\alpha(\beta+\gamma)$ बराबर है :

(A) 0 (B) $\alpha\gamma$ (C) $\beta\gamma$ (D) $\alpha\beta$

Ans. C

Sol. $\beta^2=\alpha\gamma$ so roots of the equation $\alpha x^2+2\beta x+\gamma=0$

are $\frac{-2\beta\pm2\sqrt{\beta^2-\alpha\gamma}}{2\alpha}=-\frac{\beta}{\alpha}$

This root satisfy the equation $x^2+x-1=0$

$\beta^2-\alpha\beta-\alpha^2=0$

$\Rightarrow \quad \alpha\gamma-\alpha\beta-\alpha^2=0$

$\Rightarrow \quad \alpha+\beta=\gamma$

Now, $\alpha(\beta+\gamma)=\alpha\beta+\alpha\gamma=\alpha\beta+\beta^2$

$=(\alpha+\beta)\beta=\beta\gamma$

6. Let α and β be the roots of the equation $x^2-x-1=0$. If $p_k=(\alpha)^k+(\beta)^k$, $k\geq1$, then which one of the following statements is not true ?

(A) $p_5=p_2\cdot p_3$ (B) $(p_1+p_2+p_3+p_4+p_5)=26$

(C) $p_3=p_5-p_4$ (D) $p_5=11$

माना α तथा β समीकरण $x^2-x-1=0$ के मूल हैं। यदि $p_k=(\alpha)^k+(\beta)^k$, $k\geq1$, तो निम्न में से कौनसा एक कथन सत्य **नहीं** है?

(A) $p_5 = p_2 \cdot p_3$ (B) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(C) $p_3 = p_5 - p_4$ (D) $p_5 = 11$

Ans. A

Sol. $x^2 - x - 1 = 0$ roots are λ, β.

$P_k = \alpha^k + \beta^k$

$= \alpha^{k-2}(\alpha + 1) + \beta^{k-2}(\beta + 1)$

$= (\alpha^{k-1} + \beta^{k-1}) + (\alpha^{k-2} + \beta^{k-2})$

$p_k = p_{k-1} + p_{k-2}$ .....(1)

$p_1 = \alpha + \beta = 1, \quad p_2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 3$

$p_3 = p_2 + p_1 = 4$

$p_4 = p_3 + p_2 = 7$

$p_5 = p_4 + p_3 = 11$

Now check option

7. Let $a, b \in \mathbf{R}, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to :

(A) 28 (B) 25 (C) 26 (D) 24

माना $a, b \in \mathbf{R}, a \neq 0$ इस प्रकार हैं कि समीकरण $ax^2 - 2bx + 5 = 0$ का α पुनरावृत्त मूल है, जो समीकरण $x^2 - 2bx - 10 = 0$ का भी एक मूल है। यदि β इस समीकरण का दूसरा मूल है, तो $\alpha^2 + \beta^2$ बराबर है :

(A) 28 (B) 25 (C) 26 (D) 24

Ans. B

Sol. $ax^2 - 2bx + 5 = 0 \begin{cases} \alpha \\ \alpha \end{cases}$

$\alpha = \frac{b}{a} \qquad \alpha^2 = \frac{5}{a}$

$b^2 = 5a$

$x^2 - 2bx - 10 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$\alpha + \beta = 2b \quad \alpha\beta = -10$

$\frac{b^2}{a^2} - \frac{2b^2}{a} - 10 = 0$

$b^2 - 2ab^2 - 10a^2 = 0$

$a = \frac{1}{4}$

$b^2 = \frac{5}{4}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4b^2 + 20 = 4 \times \frac{5}{4} + 20 = 25$

8. Let f(x) be a quadratic polynomial such that f(– 1) + f(2) = 0. If one of the roots of f(x) = 0 is 3, then its other root lies in :

माना f(x) एक द्विघात बहुपद है जिसके लिए f(– 1) + f(2) = 0 है। यदि f(x) = 0 का एक मूल 3 है, तो दूसरा मूल निम्न में से किस अन्तराल में स्थित है?

(A) (– 3, – 1) (B) (0, 1) (C) (1, 3) (D) (– 1, 0)

Ans. D

Sol. $f(3) = 0$

$\Rightarrow f(x) = k(x - 3)(x - \alpha)$

$f(2) + f(-1) = 0$

$k(-1)(-2 - \alpha) + k(-4)(-1 - \alpha) = 0$

$k(5\alpha + 2) = 0$

$\because k \neq 0$

$\Rightarrow \alpha = \frac{-2}{5}$

$\frac{-2}{5} \in (-1, 0)$.

9. If α and β are the roots of equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation

$2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to :

यदि α तथा β, समीकरण $x^2 + px + 2 = 0$ के मूल हैं तथा $\frac{1}{\alpha}$ तथा $\frac{1}{\beta}$, समीकरण $2x^2 + 2qx + 1 = 0$ के मूल हैं, तो $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ बराबर है :

(A) $\frac{9}{4}\left(9 - q^2\right)$ (B) $\frac{9}{4}\left(9 + p^2\right)$ (C) $\frac{9}{4}\left(9 - p^2\right)$ (D) $\frac{9}{4}\left(9 + q^2\right)$

Ans. C

Sol. $x^2 + px + 2 = 0 \begin{cases} \alpha \\ \beta \end{cases}$

$\alpha+\beta=-p$

$\alpha.\beta=2$

$$\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)\left(\alpha-\frac{1}{\alpha}\right)\left(\beta-\frac{1}{\beta}\right)$$

$$=\left(\alpha\beta+\frac{1}{\alpha\beta}+2\right)\left(\alpha\beta+\frac{1}{\alpha\beta}-\left(\frac{\beta}{\alpha}+\frac{\alpha}{\beta}\right)\right)$$

$$=\left(2+\frac{1}{2}+2\right)\left(2+\frac{1}{2}-\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}\right)$$

$$=\frac{9}{2}\left(\frac{5}{2}-\frac{p^2-4}{2}\right)=\frac{9}{4}\left(9-p^2\right)$$

10. If α and β are the roots of the equation $2x(2x+1)=1$, then β is equal to :

यदि α तथा β समीकरण $2x(2x+1)=1$ के मूल हैं, तो β बराबर है :

(A) $2\alpha(\alpha-1)$ (B) $2\alpha^2$ (C) $2\alpha(\alpha+1)$ (D) $-2\alpha(\alpha+1)$

Ans. D

Sol. $2x(2x+1)=1$

$4x^2+2x-1=0$

$4\alpha^2+2\alpha-1=0 \Rightarrow 4\alpha^2+2\alpha=1$

$$2\alpha^2+\alpha=\frac{1}{2}$$

$$\alpha+\beta=-\frac{1}{2}$$

$$\beta=-\alpha-\frac{1}{2}$$

$\beta=-\alpha-(2\alpha^2+\alpha)$

$\beta=-2\alpha-2\alpha^2=-2\alpha(\alpha+1)$

11. The integer 'k', for which the inequality $x^2-2(3k-1)x+8k^2-7>0$ is valid for every x in R, is :

(A) 0 (B) 4 (C) 2 (D) 3

पूर्णांक 'k', जिसके लिए असमिका $x^2 - 2(3k-1)x + 8k^2 - 7 > 0$, R में प्रत्येक x के लिए मान्य है, है :

Ans. D

Sol. $D < 0$

$4(3k-1)^2 - 4(8k^2 - 7) < 0$

$9k^2 - 6k + 1 - 8k^2 + 7 < 0$

$k^2 - 6k + 8 < 0$

$(k-4)(k-2) < 0$

$k \in (2, 4)$

$\Rightarrow k = 3$

12. The value of $3+\cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+.....\infty}}}}$ is equal to :

$3+\cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+.....\infty}}}}$ का मान बराबर है :

(A) $3+2\sqrt{3}$ (B) $2+\sqrt{3}$ (C) $4+\sqrt{3}$ (D) $1.5+\sqrt{3}$

Ans. D

Sol. Let $x = 3+\cfrac{1}{4+\cfrac{1}{x}}$

$\Rightarrow x = 3+\cfrac{x}{4x+1}$

$\Rightarrow x = \cfrac{13x+3}{4x+1}$

$\Rightarrow 4x^2 - 12x - 3 = 0$

$\Rightarrow x = \cfrac{12 \pm \sqrt{144+48}}{B} = \cfrac{3 \pm 2\sqrt{3}}{2}$

Since $x > 0$

so $x = 1.5 + \sqrt{3}$

13. The number of real solutions of the equation $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, is :

समीकरण $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, के वास्तविक हलों की संख्या है :

(A) 4 (B) 0 (C) 2 (D) 3

Ans. B

Sol. $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$

$3\left[\left(x+\frac{1}{x}\right)^2-2\right]-2\left(x+\frac{1}{x}\right)+5=0$

Let $x+\frac{1}{x}=t; t\in(-\infty-2]\cup[2,\infty)$

$3t^2-6-2t+5=0$

$3t^2-2t-1=0$

$t=1, t=\frac{-1}{3}$ But $t\in(-\infty,-2]\cup[2,\infty)$

$\Rightarrow$ No. real solution

14. Let $\lambda\neq 0$ be a real number. Let α,β be the roots of the equation $14x^2-31x+3\lambda=0$ and α,γ be the roots of the equation $35x^2-53x+4\lambda=0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

माना $\lambda\neq 0$ एक वास्तविक संख्या है। माना α,β समीकरण $14x^2-31x+3\lambda=0$ के मूल हैं एवं α,γ समीकरण $35x^2-53x+4\lambda=0$ के मूल है। तब $\frac{3\alpha}{\beta}$ व $\frac{4\alpha}{\gamma}$ किस समीकरण के मूल है ?

(A) $49x^2-245x+250=0$ (B) $7x^2+245x-250=0$

(C) $49x^2+245x+250=0$ (D) $7x^2-245x+250=0$

Ans. A

Sol. $14\alpha^2-31\alpha+3\lambda=0$ ...(i)

$35\alpha^2-53\alpha+4\lambda=0$ ...(ii)

$\because$ α will satisfy both give equations

$\therefore\ 14\alpha^2 - 31\alpha + 3\lambda = 0 \Rightarrow \alpha^2 - \frac{31\alpha}{14} + \frac{3\lambda}{14} = 0$ .... (iii)

$35\alpha^2 - 53\alpha + 4\lambda = 0 \Rightarrow \alpha^2 - \frac{53\alpha}{35} + \frac{4\lambda}{35} = 0$ ....(iv)

from (iii) – (iv) we get

$\lambda = 7\alpha$ put in ... (iii)

$\alpha^2 - \frac{31\alpha}{14} + \frac{3}{14}(7\alpha) = 0$

$\alpha^2 - \frac{31\alpha}{14} + \frac{3\alpha}{2} = 0$

$\Rightarrow \alpha = 0$ or $\alpha = \frac{5}{7}$

($\alpha = 0$ not acceptable as $\lambda \neq 0$)

$\therefore \alpha = \frac{5}{7}$

$\because \alpha + \beta = \frac{31}{14} \Rightarrow \beta = \frac{3}{2}$

$\because \alpha + \gamma = \frac{53}{35} \Rightarrow \gamma = \frac{4}{5}$

$\therefore \frac{3\alpha}{\beta} = \frac{10}{7}$ and $\frac{4\alpha}{\gamma} = \frac{25}{7}$

$\therefore$ Reqd. Q.E. is $\left(x - \frac{10}{7}\right)\left(x - \frac{25}{7}\right) = 0$

$(7x - 10)(7x - 25) = 0$

$49x^2 - 245x + 250 = 0$

15. The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is :

समीकरण $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$ के वास्तविक मूलों की संख्या है :

(A) 3 (B) 0 (C) 1 (D) 2

Ans. C

Sol. $\sqrt{(x-1)(x-3)} + \sqrt{(x-3)(x+3)} = \sqrt{2(2x-1)(x-3)}$

$x - 3 = 0$ and $\sqrt{x-1} + \sqrt{x+3} = \sqrt{2(2x-1)}$ .....(1)

Squaring both side; $x - 1 + x + 3 + 2\sqrt{(x-1)(x+3)} = 4x - 2$

$2\sqrt{(x-1)(x+3)} = 2x - 4$

$\sqrt{x^2+3x-x-3} = x-2$ ..... (2)

Squaring both side; $x^2 + 2x - 3 = x^2 - 4x + 4$

$6x = 7 \Rightarrow x = \frac{7}{6}$

At $x = \frac{7}{6}$ equation (2) positive = negative

Rejected

$\therefore$ real root $x = 3$

16. The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$ has :

(A) four solutions two of which are negative

(B) two solutions and only one of them is negative

(C) two solutions and both are negative

(D) no solution

समीकरण $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$:

(A) के चार हल हैं जिनमें से दो ऋणात्मक है

(B) के दो हल हैं तथा उनमें से केवल एक ऋणात्मक है

(C) के दो हल हैं जिनमें से दो ऋणात्मक है

(D) का कोई हल नहीं है

Ans. C

Sol. $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0$

Let $e^x = t$

Now $t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

Dividing equation by t^2,

$t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$

$t^2 + \frac{1}{t^2} + 8\left(t - \frac{1}{t}\right) + 13 = 0$

$\left(t - \frac{1}{t}\right)^2 + 2 + 8\left(t - \frac{1}{t}\right) + 13 = 0$

Let $t - \frac{1}{t} = z$

$z^2 + 8z + 15 = 0$

$(z+3)(z+5) = 0$

$z = -3$ ro $z = -5$

So, $t - \frac{1}{t} = -3$ or $t - \frac{1}{t} = -5$

$t^2 + 3t - 1 = 0$ or $t^2 + 5t - 1 = 0$

$t = \frac{-3 \pm \sqrt{13}}{2}$ or $t = \frac{-5 \pm \sqrt{29}}{2}$

as $t = e^x$ so t must be positive,

$t = \frac{\sqrt{13} - 3}{2}$ or $\frac{\sqrt{29} - 5}{2}$

So, $x = \ln\left(\frac{\sqrt{13} - 3}{2}\right)$ or $x = \ln\left(\frac{\sqrt{29} - 5}{2}\right)$

Hence two solution and both are negative.

17. The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1,2) and its other roots lies in the interval (2,3), is :

k के पूर्णांक मानों, जिनके लिए समीकरण $2x^2 - 8x + k = 0$ का एक मूल अंतराल (1,2) में है, तथा दूसरा मूल अंतराल (2,3) में, की संख्या है :

(A) 0 (B) 3 (C) 1 (D) 2

Ans. C

Sol. Let $f(x) = 2x^2 - 8x + k = 0$

$f(1) > 0$

$f(2) < 0$

$f(3) > 0$

$f(1) > 0$ Þ $2 - 8 + k > 0 \Rightarrow k \rangle 6$ (A)

$f(2) \langle 0 \Rightarrow 8 - 16 + k \langle 0 \Rightarrow k \langle 8$ (B)

$f(3) \rangle 0 \Rightarrow 18 - 24 + k \rangle 0 \Rightarrow k \rangle 6$ (C)

$k \in (6, 8) \therefore$ integral $k = 7$

18. The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is :

समीकरण $|x^2 - 8x + 15| - 2x + 7 = 0$ के सभी मूलों का योग है :

(A) $11 + \sqrt{3}$ (B) $11 - \sqrt{3}$ (C) $9 + \sqrt{3}$ (D) $9 - \sqrt{3}$

Ans. C

Sol. $|x^2 - 8x + 15| - 2x + 7 = 0$

$|(x - 3)(x - 5)| - 2x + 7 = 0$

Case – I : If $x \in (-\infty, 3] \cup [5, \infty)$

$x^2 - 8x + 15 - 2x + 7 = 0$

$x^2 - 10x + 22 = 0$

$x = \frac{10 \pm \sqrt{12}}{2} = 5 \pm \sqrt{3}$

$\therefore x = 5 + \sqrt{3}$ is accepted

Case – II : If $x \in (3, 5)$

$-x^2 + 8x - 15 - 2x + 7 = 0$

$x^2 - 6x + 8 = 0$

$x = 2, 4 \Rightarrow x = 4$ is only accepted

$\therefore$ Sum of roots $5 + \sqrt{3} + 4 = 9 + \sqrt{3}$

19. Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$. if $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to:

माना α, β, γ समीकरण $x^3 + bx + c = 0$ के तीन मूल है। यदि $\beta\gamma = 1 = -\alpha$, तो $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ बराबर है।

(A) 21 (B) 19 (C) $\frac{169}{8}$ (D) $\frac{155}{8}$

Ans. B

Sol. Roots of $x^3 + bx + c = 0$ are α, β, γ

$\because \beta\gamma = 1 = -\alpha$

$\therefore \alpha = -1$ ....(i)

and $\alpha + \beta + \gamma = 0$ ....(ii)

$\alpha\beta\gamma = -c$ ....(iii)

$\therefore c = 1$ ....(iv)

$\beta + \gamma = 1$ ....(v)

$\because \alpha\beta + \beta\gamma + \gamma\alpha = b$

$\Rightarrow \alpha(\beta + \gamma) + \beta\gamma = b$

$\therefore b = 0$ ....(vi)

$\therefore \beta = -\omega, \gamma = -\omega^2$ ....(vii)

$b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$

$= 0 + 2 + 3 + 6 + 8$

$= 19$

20. Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal to :

माना द्विघातीय समीकरण $x^2+\sqrt{6}x+3=0$ के मूल α, β हैं। तो $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ बराबर है :

(A) 72 (B) 81 (C) 729 (D) 9

Ans. D

Sol. $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$

Let $a_n=\alpha^n+\beta^n$

$=\frac{a_{23}+a_{14}}{a_{15}+a_{10}}$

$x^2+\sqrt{6}x+3=0 \begin{cases} \alpha \\ \beta \end{cases}$

$x^2+\sqrt{6}x+3=0 \sum_{\beta}^{\alpha}$

$x=\frac{-\sqrt{6}\pm\sqrt{-6}}{2}$

$=\sqrt{6}\left(\frac{-1\pm i}{2}\right)$

$=\sqrt{3}\left(\frac{-1\pm i}{\sqrt{2}}\right)$

$\alpha=\sqrt{3}e^{\frac{i3\pi}{4}}$

21. Let $\alpha_1,\alpha_2,.....,\alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1|\geq|\alpha_2|\geq...\geq|\alpha_7|$. Then $\alpha_1\alpha_2-\alpha_3\alpha_4+\alpha_5\alpha_6$ is equal to _____ .

माना समीकरण $x^7 + 3x^5 - 13x^3 - 15x = 0$ के मूल $\alpha_1,\alpha_2,.....,\alpha_7$ हैं तथा $|\alpha_1|\geq|\alpha_2|\geq...\geq|\alpha_7|$. है। तो $\alpha_1\alpha_2-\alpha_3\alpha_4+\alpha_5\alpha_6$ बराबर है।

Ans. 9

Sol. $x^7 + 3x^5 - 13x^3 - 15x = 0$

$\Rightarrow$ $x = 0, x^6 + 3x^4 - 13x^2 - 15 = 0$

Let $x^2 = t$

Equation $t^3 + 3t^2 - 13t - 15 = 0$

$\Rightarrow (t - 3)(t^2 + 6t + 5) = 0$

$\Rightarrow t = 3, t^2 + 6t + 5 = 0$

$\Rightarrow t = -1, -5$

$x^2 = 3 \Rightarrow x = \sqrt{3}, -\sqrt{3}$

$x^2 = -1 \Rightarrow x = i, -i$

$x^2 = -5 \Rightarrow x = -\sqrt{5}i, \sqrt{5}i$

$a_1a_2 - a_3a_4 + a_5a_6$

$= \left(\sqrt{5}i\right)\left(-\sqrt{5}i\right) - \left(\sqrt{3}\right)\left(-\sqrt{3}\right) + (i)(-i)$

$= 5 + 3 + 1 = 9$